

Morphology-matched forward readout gates for Tau Core gravity: extending SPARC residual Papers 1–3 from inverse diagnostics to predeclared family tests

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Abstract

Paper 1 found that residual-blind external structural disturbance labels are associated with increased low-acceleration residual scatter in SPARC galaxies. Paper 2 reversed the question and showed that fixed residual-shape features can infer those disturbance labels better than shuffled controls, while remaining non-unique with respect to MOND-simple and empirical RAR-like baselines. Paper 3 then built a candidate/control framework and introduced an endpoint-conditioned required- S_τ diagnostic: what preparation/readout factor would be needed to absorb a measured residual? This Paper 8 proposal turns that inverse diagnostic into a forward validation gate. The proposed gate predeclares a residual-blind observed morphology handle K_{obs} , audits whether it should be used directly or mapped to a readout-relevant proxy K_{readout} , maps K_{readout} to a Tau Core morphology-conditioned readout shell, computes a forward δg^K without using required S_τ as a predictor, and compares matched-family residual structure against wrong-family, shuffled-label, Newtonian baryonic, MOND-simple, and empirical RAR controls. The permitted claim is methodological: the morphology-matched forward-readout gate is a falsifiable, Tau-Core-specific next endpoint for the Paper 1–3 program. This paper does not claim that Tau Core has been empirically validated, that a universal galaxy law has been derived, or that the matched-family endpoint has already beaten MOND/RAR on real SPARC data.

1 Purpose and claim boundary

The first three SPARC residual papers define a cautious progression. Paper 1 asks whether external structural disturbance is associated with residual scatter [2]. Paper 2 asks whether residual-shape features can recover external disturbance labels better than shuffled labels [3]. Paper 3 turns the association into a candidate/control framework and asks what S_τ would be required to absorb the observed residual [1].

That Paper 3 S_τ diagnostic is useful but partly inverse. It asks:

given the measured endpoint, what $S_\tau(R)$ would absorb it?

The present paper defines the next gate:

residual-blind observed morphology handle K_{obs}
→ readout-relevant proxy K_{readout}
→ forward readout δg^K before endpoint scoring.

The main claim is deliberately narrow:

The Tau Core gravity bridge can sharpen Papers 1–3 by replacing inverse required- S_τ residual absorption with a morphology-matched forward-readout candidate/control gate.

This paper does not claim:

- that Tau Core is proven;
- that the morphology-matched gate has already won on real SPARC endpoints;
- that MOND, RAR, or Newtonian baryonic baselines have been superseded;
- that non-axisymmetric morphology classes can be fully tested from one-dimensional rotation curves alone;
- that the morphology families below are the final physical taxonomy.

2 From Paper 3 inverse diagnostics to forward readout

Paper 3 used an endpoint-conditioned diagnostic of the form

$$S_{\tau,\text{req}}(R) = \frac{V_{\text{obs}}(R)/V_N(R) - 1}{\alpha \ln(1 + a_0/a_N(R))}.$$

This quantity is descriptive. Because it uses V_{obs} , it cannot be treated as an independent predictor. Its correct role is to identify what kind of future nonleaky rule would be required.

The forward replacement is:

$$K_{\text{obs}} \longrightarrow K_{\text{readout}} \longrightarrow \mathcal{F}_{K_{\text{readout}}} \longrightarrow \delta g^{K_{\text{readout}}}(R) \longrightarrow \text{Score}(g_{\text{Newt}} + \delta g^{K_{\text{readout}}}).$$

Here K_{obs} is the residual-blind observed 4D morphology handle, K_{readout} is the predeclared readout-relevant morphology proxy, $\mathcal{F}_{K_{\text{readout}}}$ is the selected readout family, and $\delta g^{K_{\text{readout}}}$ is computed without using $S_{\tau,\text{req}}$, endpoint residual gain, or post-hoc family selection. The equality $K_{\text{obs}} = K_{\text{readout}}$ is allowed but not assumed; this is a Tau Core bridge consequence of treating morphology as a projection/readout structure rather than only a present-day visual class.

3 MORPHOLOGY-MATCHED-FORWARD-READOUT-GATE

For a galaxy g , let

$$K_{\text{obs},g}$$

be its residual-blind observed 4D morphology handle, and let

$$K_{\text{readout},g} \in \mathcal{K}$$

be the predeclared readout-relevant morphology proxy used for formula-shell selection. Let $\mathcal{F}_K$ denote the Tau Core readout shell assigned to readout proxy class K . Let

$$R_{g,K}$$

be the residual score after applying the forward readout family $\mathcal{F}_K$ to galaxy g under a frozen geometry and amplitude policy.

The matched-family score is

$$R_{g,\text{matched}} := R_{g,K_{\text{readout},g}}.$$

The wrong-family score is

$$R_{g,\text{wrong}} := \frac{1}{|\mathcal{K}| - 1} \sum_{K \neq K_{\text{readout},g}} R_{g,K}.$$

The central paper-level endpoint can be written as

$$\Delta_{\text{matched}} = \text{RMS}_g(R_{g,\text{wrong}}) - \text{RMS}_g(R_{g,\text{matched}}).$$

An equivalent rank endpoint is

$$\text{rank}_g(K_{\text{readout},g}) = \text{rank}(R_{g,K_{\text{readout},g}} \text{ among } \{R_{g,K} : K \in \mathcal{K}\}).$$

The gate passes only if the predeclared matched family improves residual structure more than:

1. wrong morphology families;
2. shuffled morphology labels;
3. Newtonian baryonic baseline;
4. MOND-simple baseline;
5. empirical RAR-like baseline.

This is Tau-Core-specific in a way that ordinary MOND/RAR comparisons are not: the discriminating claim is not merely lower residuals, but morphology-indexed readout specificity.

MORPHOLOGY-MATCHED-FORWARD-READOUT-GATE

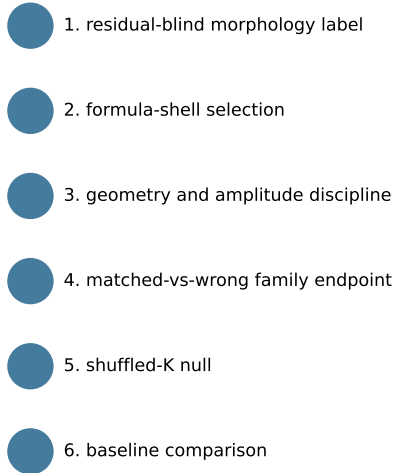
- 
1. residual-blind morphology label
 2. formula-shell selection
 3. geometry and amplitude discipline
 4. matched-vs-wrong family endpoint
 5. shuffled-K null
 6. baseline comparison

Figure 1: The proposed morphology-matched forward-readout gate. The labels and family choices must be fixed before endpoint scoring.

4 Morphology-family registry

The current gravity bridge centralizes a morphology-conditioned TGP-like readout lane. The corresponding first-pass family registry is not a universal galaxy law. It is a set of candidate readout shells whose selection must be controlled by residual-blind morphology evidence. In this paper, K_{obs} denotes the apparent 4D morphology handle and K_{readout} denotes the readout-relevant proxy used to select the shell. Thin/thick disk, bar, ring, compact, tail, and lopsided labels are therefore treated as projected 4D morphology handles unless a source-review or bridge argument promotes them to readout-relevant proxies.

Table 1: Claim-bounded morphology-family registry for the forward-readout gate.

Family	Morphology label	Formula shell	First-pass status
K_{tail}	Scale-tail spiral / diffuse LSB disk	$\sigma_{\text{morph}}(R) \sim A_K/(R + R_c)^2$	1D testable
K_{exp}	Regular exponential disk	$\sigma_{\text{morph}}(R) \sim A_K e^{-R/R_d}$	1D testable
K_{compact}	Compact finite support	$I_{\text{morph}}(R) \rightarrow I_\infty$ and $\delta g_R \sim -I_\infty/R^2$	1D testable
K_{ring}	Ring or resonance feature	localized $\sigma_{\text{morph}}(R)$ near R_{ring}	1D proxy
K_{flare}	Thick or flared disk	projected thickness/flaring kernel	1D proxy
$K_{m=2}$	Barred spiral / $m = 2$ mode	harmonic $\sigma_{\text{morph}}(R, \phi)$	velocity-field preferred
$K_{m=1}$	Lopsided / $m = 1$ mode	harmonic $\sigma_{\text{morph}}(R, \phi)$	velocity-field preferred

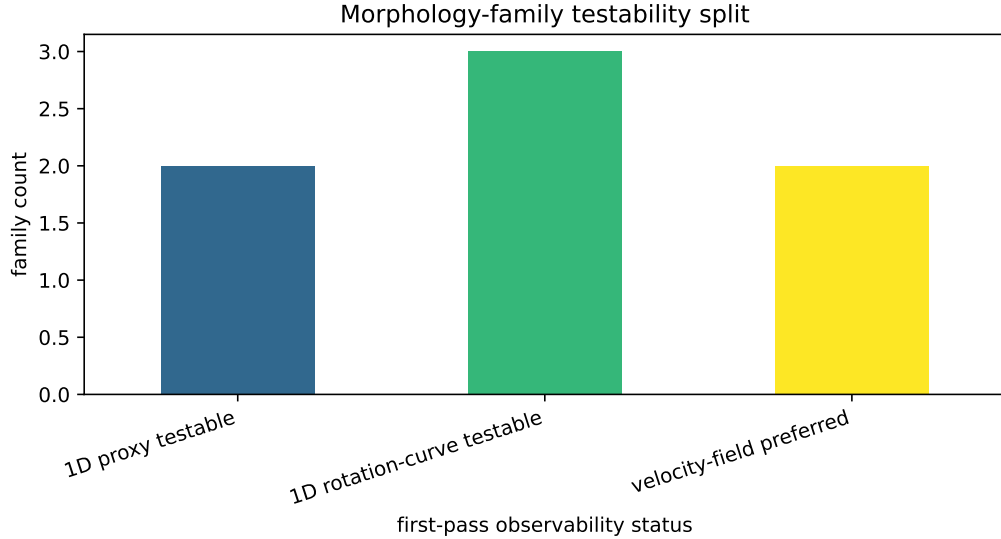


Figure 2: First-pass observability split for the current family registry. SPARC one-dimensional rotation curves are most suitable for axisymmetric or radially localized families; barred and lopsided branches require velocity-field follow-up for a full test.

5 Candidate/control continuity from Paper 3

The gate preserves the Paper 3 candidate/control discipline. It does not discard difficult controls after seeing endpoints. The same object may carry a different role under the forward gate:

candidate/control role $\rightarrow$ predeclared morphology row $\rightarrow$ matched/wrong/shuffled score.

Table 2: Paper 3 roles carried into the Paper 8 forward-gate protocol.

Galaxy	Paper 3 role	Paper 8 role
DDO126	positive anchor	matched-family candidate after residual-blind morphology labeling
DDO50	quiet control	false-positive suppression control
WLM	geometry stress	observability-caveated endpoint row
DDO168	ladder support	candidate/control consistency row
DDO154	countercontrol	wrong-family and rank-sensitivity stress row

The positive paper-level endpoint is not:

Tau Core fits every galaxy better.

The first meaningful endpoint is:

the correct predeclared morphology family ranks better than wrong families.

This is why shuffled- K nulls are mandatory. Without the shuffled-label null, a morphology-indexed family could accidentally behave like a generic flexible residual smoother.

6 Synthetic protocol fixture

This package includes a synthetic score fixture to verify the endpoint logic. It is not real SPARC evidence. It exists only so the public tests can check that the manuscript, tables, and figure pipeline agree about the intended endpoint.

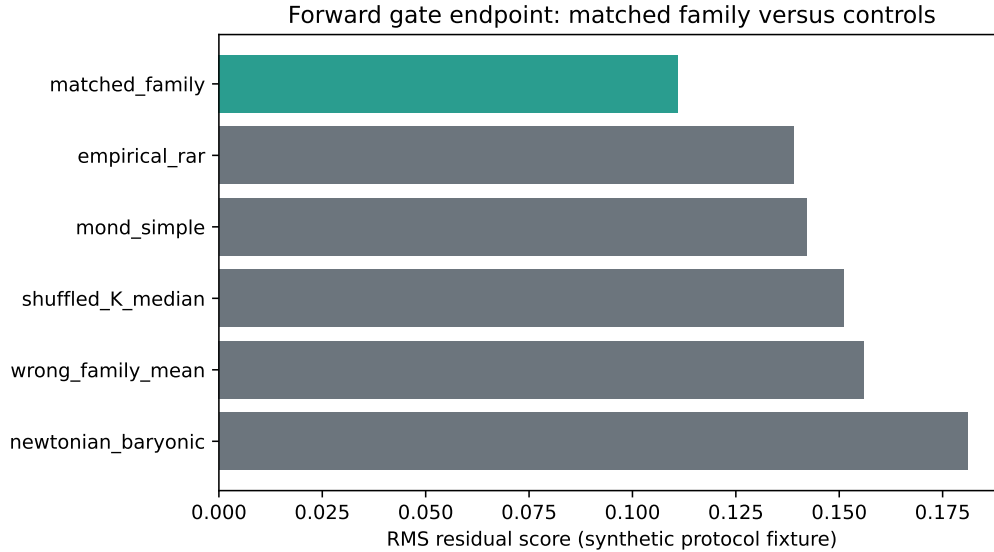


Figure 3: Synthetic protocol fixture for the endpoint schema. A real pass would require the matched-family score to beat wrong-family and shuffled-label controls, then be compared to Newtonian baryonic, MOND-simple, and empirical RAR baselines.

7 Available-data Tau-proxy preflight

The repository also includes a first available-data preflight of the matched-family idea. This is not the final Paper 8 endpoint, because the family functions are still proxy functions built from

the locally available Tau-proxy channel rather than the final morphology-specific δg^K readout shells. The preflight nevertheless tests the important next question: whether a metadata-proxy morphology family assigned before scoring performs better than wrong Tau-proxy families on held-out galaxies.

In the current 175-galaxy point-level packet, family amplitudes are fit on the train split only. On the 44-galaxy holdout split, the morphology-matched proxy beats the wrong-family mean in 0.568 of galaxies, beats the TPG/v6 comparator in 0.636 of galaxies, and beats the MOND comparator in 0.614 of galaxies. A deterministic 1000-permutation shuffled-label null gives a holdout null mean of 0.0133 for the mean matched-minus-wrong endpoint, compared with the observed value -0.0326 , with $p \simeq 0.264$ for shuffled labels being at least as good. Thus the available proxy signal is positive but not yet a discovery-level morphology-specific result. These numbers are recorded as a preparation result, not as empirical validation of Tau Core. They motivate replacing the generic Tau proxy with source-native morphology-family readout shells and a stronger shuffled-label null in the final endpoint.

A second preflight replaces the single available Tau-proxy channel with different radial formula-shell proxies assigned by morphology class. This is closer in form to the intended Paper 8 family test, but the current available-data version is weaker: on the same 44-galaxy holdout split it beats the wrong-shell mean in 0.500 of galaxies, beats TPG/v6 in 0.477 of galaxies, and has a shuffled-null $p \simeq 0.479$ for the mean matched-minus-wrong endpoint. This negative preflight result is useful. It indicates that merely assigning naive radial shell shapes is not sufficient; the final endpoint needs source-native morphology labels, dimensionally audited 4D readout shells, and a nonleaky amplitude policy.

The source-native bridge formula preflight then consumes a declared morphology-parameter manifest and evaluates the concrete morphology formulas as δv_K^2 readout kernels: the scale-tail $n = 2$ branch, the finite exponential disk Freeman/Bessel kernel, the compact finite-source exterior response, and the thick/flared damped vertical-kernel response. With the same available-data morphology labels and train-only amplitudes, the 44-galaxy holdout split gives a much stronger morphology-specific endpoint: the matched bridge formula beats the wrong-formula mean in 0.886 of galaxies, with a 1000-permutation shuffled-label null giving $p \simeq 0.002$ for the mean matched-minus-wrong endpoint. However, it beats TPG/v6 and MOND in only 0.477 of holdout galaxies each, and its mean matched-minus-baseline residual is still worse than both baselines. Thus the bridge-formula result supports the Paper 8 morphology-family specificity hypothesis, but it is not an empirical claim that Tau Core outperforms TPG/MOND.

A confidence diagnostic on the manifest shows why the baseline claim remains blocked. Restricting the holdout set to confidence ≥ 0.75 preserves the matched-vs-wrong signal, with matched formulas beating wrong formulas in 0.906 of galaxies, but the TPG/MOND comparison is still mixed. Removing low-inclination or large-distance-error objects improves some baseline comparisons, so data quality matters. It is not the whole bottleneck: the remaining blockers are source-native morphology-scale extraction, amplitude policy, and normalization of the bridge formula kernels.

An amplitude-policy diagnostic supports the same conclusion. Keeping the source-native bridge kernels fixed, the family-unconstrained policy gives the strongest matched-vs-wrong signal but remains weak against TPG/MOND baselines. A simple family-to-global shrinkage policy preserves most of the matched-vs-wrong signal, 0.864 on holdout, while raising the MOND win fraction to 0.636 and reducing the mean MOND gap. This is not a selected endpoint claim; it identifies amplitude normalization as a required Tau-side modeling gate before any baseline-superiority claim is allowed.

A shrinkage-path scan makes this more precise. Varying the family amplitude weight from 0 to 1 shows that the current best compromise lies near family weight 0.4–0.5. At weight 0.5, the

holdout matched-vs-wrong fraction is 0.864 and the MOND win fraction is 0.636, while TPG/v6 remains at 0.477. This identifies the normalization target for the next Tau-side gate: a physical source-normalization rule should recover a partially pooled family amplitude, not a fully free family amplitude chosen only for endpoint fit.

A train-selected shrinkage diagnostic reduces the post-hoc risk in that conclusion. Four selection rules using train split metrics only all select family weight 0.40. Evaluated on the 44-galaxy holdout split, this train-selected policy keeps matched-vs-wrong specificity at 0.818, beats MOND in 0.614 of galaxies, and remains mixed against TPG/v6 at 0.477. This supports a stable partially pooled normalization range, but it is still not a validated amplitude-normalization law or a baseline-superiority endpoint.

A family-breakdown diagnostic then asks where this signal lives. Under the train-selected 0.40 shrinkage policy, the holdout matched-vs-wrong specificity is strongest for the scale-tail spiral and thick/flared families, both at 0.900. The thick/flared row is also competitive against both TPG/v6 and MOND in this diagnostic, while the scale-tail row remains MOND-competitive but TPG-blocked. The exponential-disk and compact finite-source rows are weaker under the current available-data manifest. This is useful as a failure map: it points to family-specific source normalization and morphology-observable extraction as the next modeling gates, not to a validated family-level empirical claim.

An observable-quality diagnostic separates part of that failure map. Joining the train-selected scores to the residual-blind morphology manifest shows that, on holdout, the thick/flared family is the cleanest current-best-case row, while the compact finite-source, exponential-disk, and scale-tail rows are quality-limited under the available-data proxies. In the clean holdout manifest subset, matched formulas beat wrong formulas in 0.900 of galaxies and beat MOND in 0.750, whereas quality-caveated rows fall to 0.750 and 0.500. This does not rescue the endpoint by filtering after the fact; it identifies residual-blind morphology-observable extraction as a required predeclared gate for the next run.

A predeclared quality-gate diagnostic makes that next run more concrete. Under the fixed train-selected shrinkage policy, the clean-manifest gate and the confidence- ≥ 0.75 -and-clean gate each retain 20 holdout galaxies across all four formula families, with matched-vs-wrong specificity 0.900, TPG/v6 win fraction 0.500, and MOND win fraction 0.750. A broader no-large-distance-error gate retains 26 holdout galaxies and remains candidate-level, with matched-vs-wrong 0.885, TPG/v6 0.538, and MOND 0.731. These are not selected discovery endpoints; they are candidate observability rules that must be declared before endpoint scoring in a future Paper 8 run.

The corresponding shuffled-label null reveals the tradeoff. On holdout, the full all-galaxy gate has the strongest shuffled-null support for morphology specificity, with $p \simeq 0.036$ for the beats-wrong fraction and $p \simeq 0.013$ for the mean matched-minus-wrong endpoint. The no-low-inclination gate gives similar values, $p \simeq 0.041$ and $p \simeq 0.011$. Cleaner gates improve the MOND/TPG comparison but have weaker shuffled-null support because the sample shrinks. Thus Paper 8 should not silently choose the cleanest-looking subset; it should predeclare whether the endpoint prioritizes observability-clean baseline competitiveness or larger-sample morphology-label specificity.

Combining these diagnostics gives a concrete preparation-state endpoint decision matrix. The current natural primary endpoint candidate is the no-low-inclination gate: it retains 35 holdout galaxies, remains baseline-competitive, and has strong shuffled-null support. The full all-galaxy gate should remain a fuller-sample morphology-specificity support lane, while the no-large-distance-error gate is the strongest secondary baseline-competitiveness lane. These roles must be predeclared before any future endpoint scoring; they are not retrospective proof of Tau Core.

The repository therefore records a predeclared endpoint protocol sheet. It freezes the current preparation-state choices: primary lane no-low-inclination, fuller-sample support lane all, secondary

baseline lane no-large-distance-error, train-selected family-to-global shrinkage with weight 0.40, residual-blind morphology-family assignment, matched-vs-wrong plus shuffled-family null specificity metrics, and TPG/v6 plus MOND comparator metrics. It also records forbidden inputs, including endpoint residual gain, required- S_τ , post-hoc gate choice, and per-galaxy amplitude tuning. This is the guardrail for the next run, not a claim that the present diagnostic run validates Tau Core.

8 Minimal numerical protocol

The minimal public endpoint protocol is:

1. Assign an external or residual-blind observed 4D morphology handle $K_{\text{obs},g}$ to each galaxy.
2. Predeclare whether $K_{\text{obs},g}$ is used directly or mapped to a readout-relevant proxy $K_{\text{readout},g}$ using only residual-blind source evidence.
3. Select a formula shell $\mathcal{F}_{K_{\text{readout},g}}$ from the predeclared registry.
4. Fix geometry parameters from morphology, not from residual endpoints.
5. Use either a frozen amplitude rule or a tightly bounded amplitude fit.
6. Compute the matched-family score.
7. Compute wrong-family scores for all other candidate families.
8. Run shuffled- K nulls.
9. Compare against Newtonian baryonic, MOND-simple, and empirical RAR baselines.

For SPARC-only first-pass work, the cleanest initial family set is:

$$\{K_{\text{tail}}, K_{\text{exp}}, K_{\text{compact}}, K_{\text{ring}}, K_{\text{flare}}\}.$$

The $m = 1$ and $m = 2$ branches should be treated as caveated proxy rows in one-dimensional rotation curves and as real tests only when velocity-field products are available.

9 Why this would be a stronger Tau Core test

The new gate is stronger than a residual association because it creates a family-selection burden. A theory with morphology-indexed readout families should not only reduce residuals; it should reduce them more when the correct morphology family is used than when a wrong family is used. MOND-simple and empirical RAR baselines are strong comparators, but they do not naturally provide a predeclared morphology-family rank test of this kind.

The result that would matter is therefore:

$$\Delta_{\text{matched}} > 0$$

with a rank distribution in which $K_{\text{readout},g}$ is systematically near the top, under residual-blind observed morphology handles, predeclared readout-proxy rules, and shuffled-label controls. If this fails, the bridge loses specificity. If it passes, the Paper 1–3 residual-disturbance program becomes a forward-predictive morphology-readout program.

10 Readiness and remaining blockers

Table 3: Current readiness of the Paper 8 forward endpoint.

Layer	Status
Paper 1 residual-blind disturbance association	available
Paper 2 residual-shape inference and shuffled-label null	available
Paper 3 candidate/control ladder and required- S_r diagnostic	available
Morphology-family formula registry	defined in this package
Available-data matched Tau-proxy preflight	run; proxy only
Source-native formula preflight	diagnostic-ready
Train-selected shrinkage policy	diagnostic-ready
Quality-gate shuffled-null controls	diagnostic-ready
Endpoint decision matrix	predeclaration-ready
Predeclared endpoint protocol	ready as guardrail
Accepted source-native component tables for full sample	pending
Residual-blind morphology labels for full endpoint sample	pending
Empirical discovery claim	blocked

Thus the present package is preparation-ready, not discovery-ready. Its next scientific upgrade is not to strengthen the claim language, but to replace available-data proxy morphology observables with accepted residual-blind morphology inputs and then run the predeclared endpoint protocol without changing gates after seeing the scores.

For that reason the repository also includes a morphology-observable intake schema. The schema records the fields required for the next run: residual-blind family labels, confidence and caveat fields, inclination and distance-quality observables, family-specific scale/support/thickness parameters, provenance, and explicit forbidden sources. In particular, it forbids using endpoint residual gain, required- S_r , endpoint-selected radii, post-hoc family choice, or per-galaxy residual tuning as morphology inputs.

A gap audit compares the proxy manifest against this intake schema. It shows that the present package is coverage-rich but acceptance-limited: active families have broad proxy coverage, while the kernel-driving observables still need accepted residual-blind morphology sources before a discovery-style endpoint can be claimed. This makes the next upgrade concrete. The task is not to alter the endpoint gates after seeing scores, but to replace proxy morphology fields by pre-scoring observables and rerun the frozen protocol.

The repository therefore records a morphology-observable source-upgrade plan. This plan prioritizes P0 family-label, confidence, caveat, and provenance fields, then P1 active-family kernel observables such as scale, tail, compact-support, and thickness parameters. Optional ring, bar, and lopsided branches remain caveated unless external morphology or velocity-field support is supplied before scoring. The plan is a source-collection protocol, not a claim that the accepted sources have already been assembled.

Finally, the repository includes an accepted-observable manifest template and validator. The template is intentionally endpoint-blocked: it keeps the galaxy rows and pre-scoring geometry fields, but leaves accepted morphology labels, kernel observables, and provenance to be supplied from residual-blind sources. This prevents the current proxy manifest from being silently promoted into a discovery input.

An accepted-manifest readiness gate then lifts the field-level validation to an endpoint-level decision. For the current empty template, the gate returns `BLOCKED_ACCEPTED_OBSERVABLES_MISSING`. A future populated manifest must pass this gate before the frozen blind endpoint protocol is run.

A final frozen-endpoint launch guard records the same boundary operationally. In the current package it returns `LAUNCH_BLOCKED` with endpoint scores not computed. Thus the package preserves a complete path to a future blind endpoint while preventing the proxy manifest or empty accepted template from being used as discovery evidence.

The package also records an external morphology source registry for the missing accepted inputs. SPARC remains the sample and baryonic/rotation baseline; S4G is the first morphology/decomposition source; NED/NED-D supplies identity, distance, and provenance checks; Dust-Pedia is a multiband fallback/validation layer; and PHANGS is reserved for optional high-quality non-axisymmetric or velocity-field branches. The associated SPARC crossmatch table is a to-be-filled acquisition template, not accepted-source coverage.

The first acquisition pass is included in the reproducibility package. It downloads SPARC Table 1 from the official SPARC site and S4G Pipeline 4 tables from VizieR/CDS, yielding 175 SPARC master rows, 77 S4G crossmatches, and 75 S4G/SPARC-derived disk-scale candidates. These candidates can support future source-native scale-radius fields, but they do not complete the accepted manifest: family labels, tail/ring/thickness observables, caveats, and full provenance still require the accepted-source audit path.

The repository now also builds a partial accepted morphology-observable manifest from this acquisition layer. The manifest promotes 75 S4G/SPARC-derived scale-radius observables to field-level accepted-source status, while keeping all 175 rows endpoint-blocked. This is intentional: proxy family labels still require external morphology-label audit, and scale-tail, compact, and thick/flared branches still require their family-specific source-native kernel observables. Thus the accepted manifest draft is a data-readiness upgrade, not an endpoint score.

An accepted-manifest audit then separates audit lanes. Its closest near-term lane contains 13 exponential-disk rows with accepted S4G/SPARC scale radius and no remaining kernel-field blocker beyond the external family-label audit. These rows are not endpoint-ready; they are the next residual-blind morphology-label audit pool.

A targeted S4G component-label audit strengthens that pool. All 13 rows receive external disk-family support from S4G component decompositions; 6 rows have strict `D:expdisk` support and 7 rows remain caveated because the S4G support is barred or edge-disk. This creates a first narrow dry-run candidate lane, but endpoint scores are still not computed in the present package.

For diagnostic purposes, the package also runs an explicitly labeled narrow dry-run on this S4G-supported exponential-disk pool. The dry-run applies the accepted-scale Freeman/Bessel exponential-disk shell and records a frozen global train amplitude, shrinkage amplitudes, pool-fit diagnostic amplitudes, and a leave-one-galaxy-out all-13 stability policy. The outcome is mixed rather than confirmatory: on the 6 strict rows, the all-13 pool-fit amplitude beats TPG/v6 in 4/6 cases but beats MOND in only 2/6 cases, while the frozen global train amplitude beats TPG/v6 in 1/6 cases. The leave-one-galaxy-out all-13 policy beats TPG/v6 in 3/6 strict cases and MOND in 2/6 strict cases. This is a useful executable sanity check, not a validation result or frozen endpoint claim.

A final failure/sensitivity audit varies only the accepted scale radius by fixed multipliers 0.75, 1.0, and 1.25 under the leave-one-galaxy-out amplitude policy. The strict lane remains mixed across these multipliers, indicating that the current failure mode is unlikely to be solved by a single radius rescaling alone. The next refinement should therefore test morphology-subtype splitting or readout normalization rather than treating the scale radius as the only missing ingredient.

The package also records the corresponding inverse diagnostic: which morphology family would be inferred if the rotation-curve score table itself selected the readout family. This diagnostic is not residual-blind and is not allowed to define the accepted Paper 8 morphology labels. In the current run, the rotation-inferred family matches the predeclared proxy family in about 0.354 of rows,

and matches the external S4G-supported exponential-disk label in about 0.308 of the 13 audited rows. This low agreement is not a validation result; it is a hypothesis-generation signal that the current external morphology labels and the rotation-preferred readout families may require subtype splitting.

The same diagnostic motivates a more cautious morphology-memory/history layer. Because the Tau Core bridge treats morphology as a readout structure rather than only a present-day visual class, the currently observed galaxy shape may be an insufficient proxy for the readout-relevant morphology: the galaxy may not always have had its current structure, and a solved 4D readout may encode integrated or delayed morphology/history information. The repository therefore adds a morphological memory/history proxy table. It combines source-side caveats with the inverse readout-family diagnostic and marks possible current-shape/readout-history mismatches. In the current run, the current proxy family and rotation-inferred family disagree in 113/175 rows, and 9/13 S4G-supported exponential-disk rows do not infer the exponential-disk readout family. This layer is not an accepted morphology label, not an endpoint score, and not evidence that a historical morphology has been recovered; it is a pre-endpoint hypothesis layer for future residual-blind history or morphology-memory observables.

To make that next data step operational, the repository also constructs a morphology inspection queue. The queue ranks galaxies for residual-blind image, decomposition, HI, projection, and velocity-field review, and records the requested external observables rather than changing any endpoint label. In the current run it yields 4 P0 and 18 P1 inspection targets. The top P0 cases are NGC0300, NGC6503, NGC0100, and NGC0247; all are S4G-matched current exponential-disk cases whose diagnostic readout preference points toward a scale-tail or projection-sensitive memory/history check. This queue is an acquisition plan only, not validation and not an accepted morphology manifest.

The four P0 rows are then converted into residual-blind inspection packets. Each packet records the current source context, the requested external observables, suggested catalogue lookup links, blank review fields, and forbidden inputs. The common requested observables across all four P0 packets are residual-blind multiband morphology labels, deep optical or infrared outer-disk profiles, and HI extent or asymmetry checks; NGC0247 additionally requests bar-length and velocity-field support. These packets are manual source-collection templates, not filled morphology labels.

Finally, the P0 packet layer is translated into an external imaging request manifest. This manifest records S4G-derived sky coordinates, suggested fields of view, and residual-blind source URLs for DSS2, 2MASS, WISE, NED, and SIMBAD lookups. The recommended fields of view are 8.0 arcmin for NGC0100, 29.745 arcmin for NGC0247, 20.288 arcmin for NGC0300, and 10.030 arcmin for NGC6503. This makes the next image-based morphology review operational while still leaving classification and history/memory judgments blank.

For practical review, the package also renders these four requests into a local HTML imaging-review dashboard. The dashboard is an offline launch page with per-galaxy source links, requested-observable chips, and blank review checklists. It does not embed a classification result or an accepted label; it only makes the residual-blind morphology review executable by a human reviewer.

A final source-availability preflight queries SkyView for DSS2 Red, 2MASS-K, and WISE W1 availability at the requested P0 coordinates and fields of view. All 12 P0 survey requests return at least one image candidate in the current run: 4/4 for DSS2 Red, 4/4 for 2MASS-K, and 4/4 for WISE W1. The audit intentionally records availability counts and stable query URLs only, not temporary FITS URLs and not image classifications.

The package then renders these available SkyView sources into 12 local PNG preview panels, one for each P0 galaxy–survey pair. All previews render successfully at 300 by 300 pixels. These previews are included only as review source material for a future residual-blind human morphology

audit; no automated image classification, accepted label, or endpoint score is emitted.

The rendered panels are then assembled into a residual-blind visual review template. In this layer all review fields remain unfilled placeholders, and the package records the field schema needed for a future accepted morphology manifest: present-day morphology, outer-disk/LSB/tail evidence, HI asymmetry evidence, bar or $m = 2$ support, projection caveats, vertical/flare/warp evidence, compact/bulge support, ring/resonance evidence, a morphological memory/history proxy judgment, reviewer confidence, source list, and a residual-blind family recommendation. The template is not yet an accepted manifest; endpoint residual gains, required- S_τ diagnostics, best-fit readout families, MOND/RAR/TGP comparison scores, and post-hoc family switching remain forbidden inputs.

A completion gate then checks whether this visual review template has actually been filled before any accepted-manifest promotion is allowed. In the current package the gate returns `BLOCKED_VISUAL_REVIEW_PENDING`: all four P0 rows remain blocked, all 60 reviewer fields are still placeholders, accepted-manifest promotion is disallowed, and endpoint scores are not computed. This is a protocol safeguard, not a negative empirical result.

To make the blocked state actionable, the package also emits a P0 visual review handoff. The handoff lists the required residual-blind fields, local preview panels, allowed source types, and forbidden endpoint-derived inputs for each P0 galaxy. Its status is `READY_FOR_RESIDUAL_BLIND_HUMAN_REVIEW`, but it still does not promote labels or compute endpoint scores.

The handoff is finally converted into a visual-review response intake template and validator. A Codex/source-reviewed P0 response then fills the four response rows from residual-blind source evidence: S4G Pipeline 4 components and scales, SPARC HI mass/radius evidence, DustPedia/VizieR source matches, PHANGS public-sample coverage checks, and NED/SIMBAD/SkyView lookup material. This is not a human visual review and not an endpoint calculation. It records that forbidden endpoint-derived inputs are absent: no residual gain, required- S_τ diagnostic, best-fit Tau Core readout family, MOND/RAR/TGP comparison score, or post-hoc family switching is used as a label input. The validator therefore returns `READY_FOR_INDEPENDENT_ACCEPTED_MANIFEST_AUDIT` for all four P0 rows.

A response-to-manifest promotion gate records the same boundary at the next step. For the Codex/source-reviewed P0 response it also returns `READY_FOR_INDEPENDENT_ACCEPTED_MANIFEST_AUDIT`. This does not run a frozen endpoint and does not validate Tau Core; it only allows the four P0 rows to enter a source-review audit lane. The package therefore builds a P0 Codex-source-reviewed label manifest with decision `P0_CODEX_SOURCE_REVIEW_LABELS_CREATED_NOT_ENDPOINT`. All four rows receive the residual-blind family recommendation `K_exponential_disk`, with caveats for edge projection, bar structure, or nuclear components where applicable. No full 175-galaxy endpoint labels are created and no endpoint scores are computed.

The bridge interpretation then separates this apparent 4D morphology handle from a readout-relevant morphology proxy. Thin/thick disk, bar, ring, compact, and tail labels are treated as projected 4D morphology handles rather than proven fundamental Tau-side classes. In notation, the source-review layer distinguishes K_{obs} from K_{readout} , and the formula shell is selected as $\mathcal{F}_{K_{\text{readout}}}$, not automatically as $\mathcal{F}_{K_{\text{obs}}}$. In the P0 set, the apparent 4D handle is `K_exponential_disk` for all four rows, but source-side caveats split the readout-relevant proxy layer into a clean exponential-disk control, a projection-corrected expdisk candidate, a barred/ $m = 2$ overlay candidate, and a compact-core overlay candidate. The emitted proxy table carries this distinction through `k_obs`, `k_readout`, `readout_proxy_source`, `promotion_status`, and `formula_shell`. This layer uses no rotation residuals and computes no endpoint scores.

A narrow P0 pilot then consumes only these four Codex/source-reviewed labels. Under the primary leave-one-galaxy-out all13 exponential-disk policy, the Tau readout beats TPG/v6 in 0.75

of the P0 cases and beats MOND in 0.25 of the P0 cases. The formula-shell proxy slice is stronger against TPG/v6 on this small P0 set, beating the wrong-family mean and TPG/v6 in 1.0 of cases, while still beating MOND in only 0.25 of cases. The score table records `scored_formula_shell`; in this narrow pilot it remains the direct `K_exponential_disk` control, while three P0 rows are flagged `readout_proxy_overlay_not_scored=True` because their projection, barred/ $m = 2$, or compact-core readout-proxy shell has not yet been scored. This is a useful pilot signal for the TPG-like readout direction, but it is not the frozen 175-galaxy endpoint and not empirical validation.

The package then turns the open review fields into a concrete source-acquisition plan using the requested source families: S4G, NED/NED-D, DustPedia, HI surveys, and PHANGS. This plan maps each missing P0 review field to source families, including HI extent/asymmetry checks for outer-disk and morphology-memory/history support and PHANGS/S4G support for optional bar or velocity-field evidence. All tasks remain `TO_BE_ACQUIRED_RESIDUAL_BLIND`; the layer is not an accepted morphology manifest and computes no endpoint scores.

The requested source families are then queried directly where an automated source path exists. DustPedia direct matches are found for NGC0300 only in the queried VizieR tables; PHANGS public sample coverage is not found for the four P0 galaxies, including NGC0247; and SPARC HI mass/radius evidence is present for all four P0 galaxies. These source records are useful review inputs, but they do not classify morphology and do not create accepted labels.

The package also creates a source-assisted review response draft. This draft fills the P0 response fields with residual-blind source evidence and explicit blockers, including the PHANGS no-coverage blocker for NGC0247. It remains blocked because a human residual-blind review and independent accepted-manifest audit are still missing. It is not an accepted morphology manifest and not an endpoint score.

The next source-side test audits requested source-family availability. In the current package, S4G is partially source-ready for the P0 galaxies through existing crossmatches and disk-scale candidates, NED/NED-D lookup paths are ready, DustPedia is directly matched only for NGC0300, HI mass/radius evidence is ready for all four P0 galaxies through SPARC, and PHANGS public sample coverage is not found for the four P0 galaxies. This availability audit does not classify morphology and does not compute endpoint scores.

The package closes this preparation chain with a consolidated P0 review-pipeline status dashboard. The dashboard reports thirteen stages and the current overall decision `P0_CODEX_SOURCE_REVIEW_LABELS_READY_FULL_ENDPOINT_BLOCKED`: the P0 Codex/source-reviewed audit-label lane is populated, while full endpoint labels and endpoint scoring remain disabled.

11 Conclusion

Paper 3 is the natural launching point for this forward gate because it already contains the candidate/control framework and the required- S_τ diagnostic. The present contribution is to state the next gate in a falsifiable form:

$$\begin{aligned}
&\text{observed morphology handle } K_{\text{obs}} \\
&\rightarrow \text{readout proxy } K_{\text{readout}} \\
&\rightarrow \text{formula shell } \mathcal{F}_{K_{\text{readout}}} \\
&\rightarrow \delta g^{K_{\text{readout}}} \rightarrow \text{matched vs wrong family endpoint.}
\end{aligned}$$

The claim-safe breakthrough target is not a universal fit claim. It is a morphology-specific prediction claim: the correct Tau Core morphology readout family, chosen before endpoint scoring,

should improve residual structure more than wrong morphology families and shuffled morphology labels, while remaining competitive against Newtonian baryonic, MOND-simple, and empirical RAR baselines.

That is the first clean endpoint where the Tau Core bridge can become more than residual commentary.

References

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